Visualization

Durjoy Sarker

Table of contents

Barplot

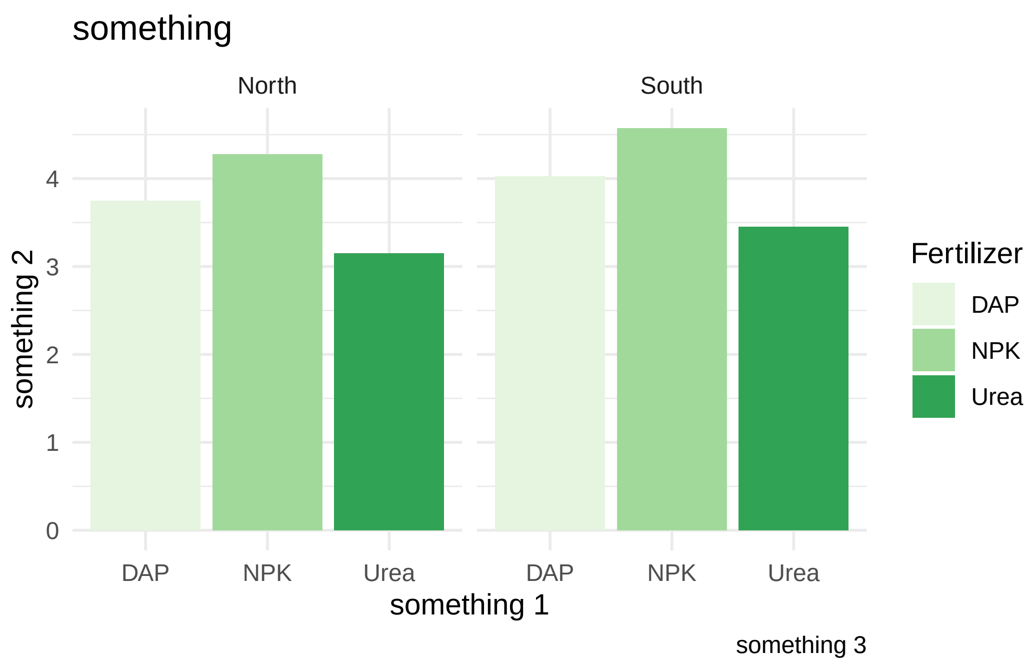
```
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
data <-  read.csv("G:/Lesson/Done/Research/R Language/R-Language/Self-Learning/ggplot/agricu
ggplot(data, aes(Fertilizer,Yield,fill=Fertilizer)) +
  geom_bar(stat = "summary" )+
  facet_wrap(~Region)+ #very very useful function. it separate the plots for each region.
  theme_minimal()+
  labs(title = "something",x="something 1",y="something 2",caption="something 3") +
  scale_fill_brewer(palette = "set7") #professional and readable plots for reports.
```

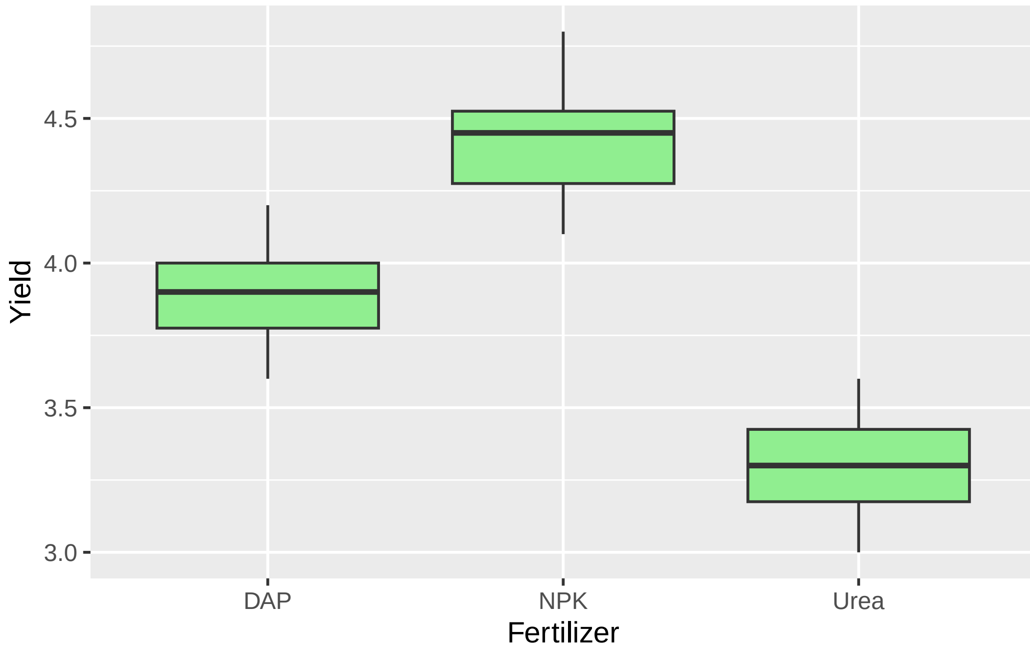
```
Warning: Unknown palette: "set7"
```

```
No summary function supplied, defaulting to `mean_se()`
No summary function supplied, defaulting to `mean_se()`
```



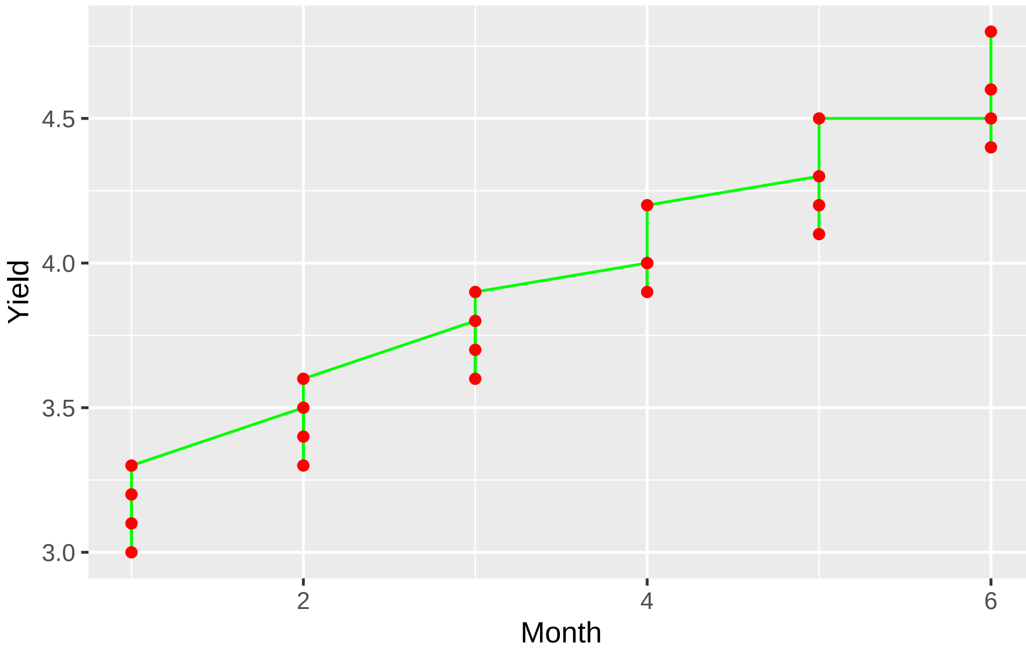
Boxplot

```
ggplot(data, aes(Fertilizer,Yield))+
geom_boxplot(fill= "lightgreen")
```



Line and Point

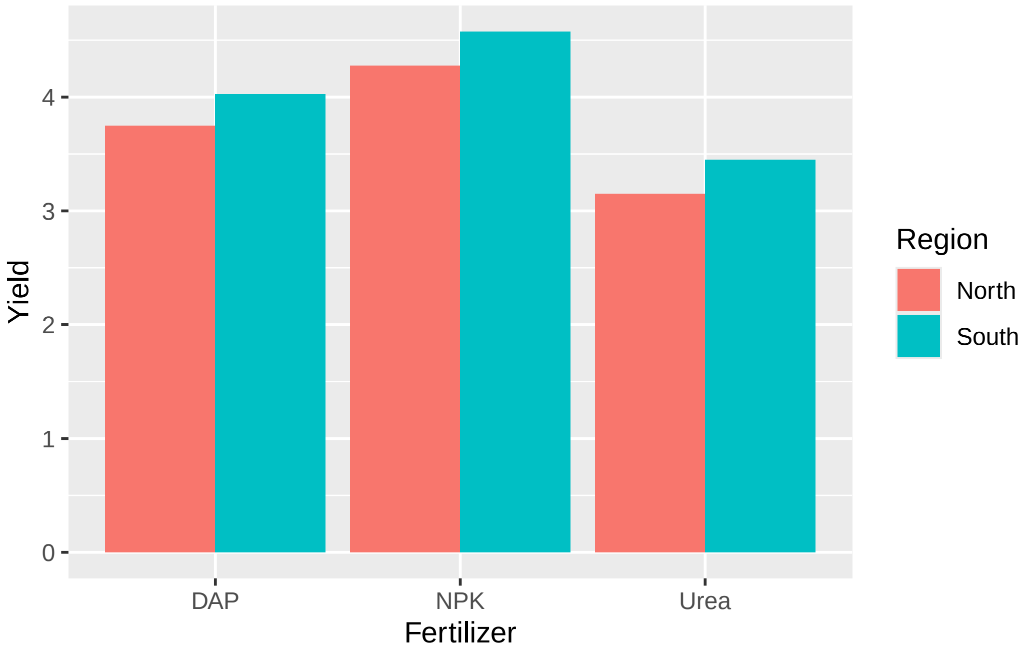
```
ggplot(data, aes(Month,Yield))+  #for better result the two variable should be numerical
  geom_line(col="green")+
  geom_point(col="red")
```



position = "dodge" and the facet_warp difference

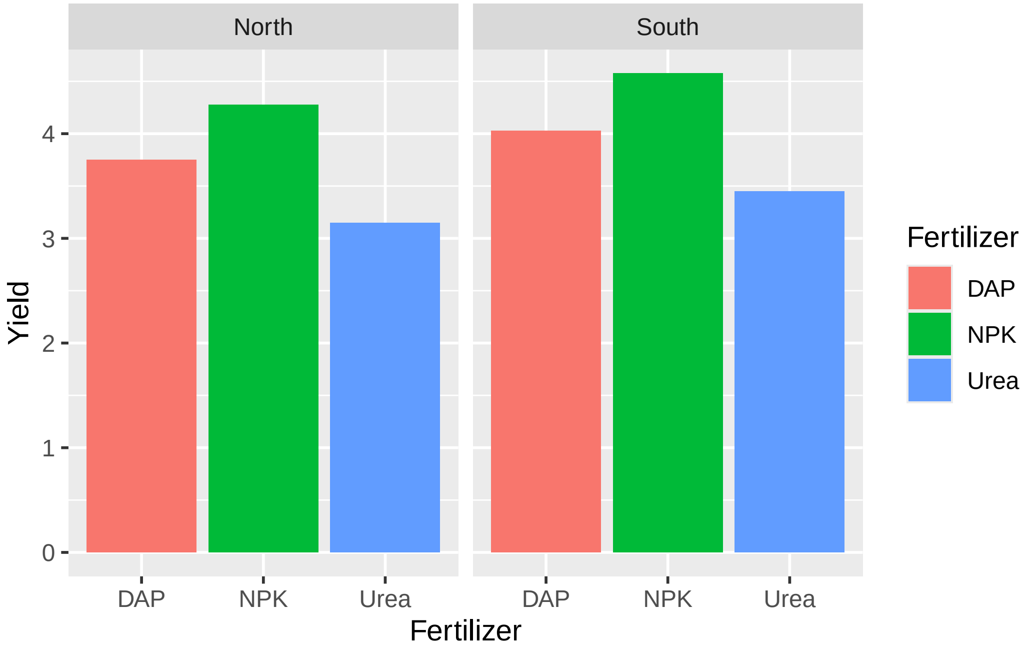
```
ggplot(data, aes(Fertilizer,Yield,fill= Region)) +
  geom_bar(stat = "summary",position = "dodge" )
```

```
No summary function supplied, defaulting to `mean_se()`
```




```
ggplot(data, aes(Fertilizer,Yield,fill= Fertilizer)) +
  geom_bar(stat = "summary")+
  facet_wrap(~Region)
```

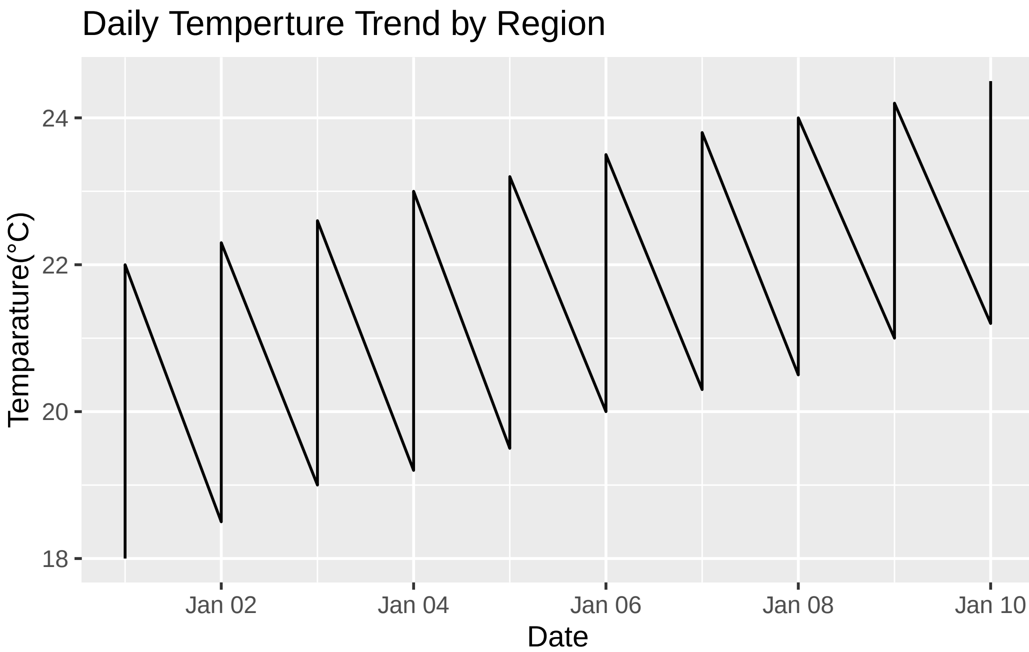
```
No summary function supplied, defaulting to `mean_se()`
No summary function supplied, defaulting to `mean_se()`
```



Time-series visualization

```
data1 <- read.csv("G:/Lesson/Done/Research/R Language/R-Language/Self-Learning/ggplot/time_se

data1$Date <- as.Date(data1$Date) #this is so much important it convert the text into date
ggplot(data1,aes(Date,Temperature), colour=Region)+
  geom_line() +
  labs(title= "Daily Temperture Trend by Region", x= "Date", y= "Temparature(°C)")
```



Grouping and ploting

Had to add caption(____)

```
#First start grouping the data on based on fertilizer
library(dplyr)
```

```
Warning: package 'dplyr' was built under R version 4.4.3
```

```

Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

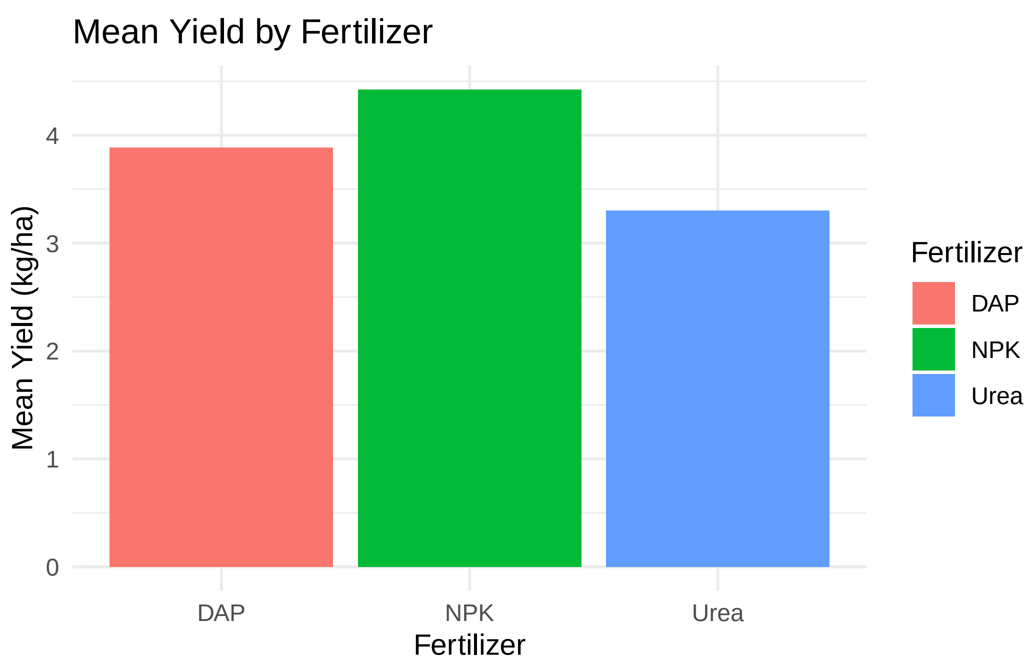
```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
summary_data <- data %>%
  group_by(Fertilizer) %>%
  summarise(
    mean_yield = mean(Yield, na.rm= TRUE),
    median_yield = median(Yield, na.rm= TRUE),
    sd_yield = sd(Yield, na.rm = TRUE)
    )

#bar plot of mean yield by fertilizer
ggplot(summary_data, aes(Fertilizer, mean_yield, fill = Fertilizer))+
  geom_bar(stat="summary")+
  labs(title = "Mean Yield by Fertilizer", x = "Fertilizer", y = "Mean Yield (kg/ha)") +
  theme_minimal()
```

```
No summary function supplied, defaulting to `mean_se()`
```



Grouping based on the multiple column + Visualization

```r
#Bar Plot of Mean Yield by Fertilizer and Region
sad_multi <- data %>%
```

```
  group_by(Fertilizer,Region) %>%
  summarise(
    yield_mean = mean(Yield, na.rm=TRUE)
    )
```

```
`summarise()` has grouped output by 'Fertilizer'. You can override using the
`.groups` argument.
```

```
ggplot(sad_multi,aes(Fertilizer,yield_mean,fill=Region))+
  geom_bar(stat="summary", position = "dodge") +
  labs(title = "Mean Yield by Fertilizer and Region", x = "Fertilizer", y = "Mean Yield (kg/h
  theme_minimal()
```

```
No summary function supplied, defaulting to `mean_se()`
```